Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Sets up enumerated population using valid numbering stating range used (PI)	AO2.4	E1	Give each student a number from (000)1 to 3200 or equivalent
Explains how to obtain sample with respect to a specified range of random numbers. Accept random number generator / calculator set to give numbers from 1 to 3200 Do not accept 'drawn from a hat' (impractical for 3200 population)	AO2.4	E1	Generate random four digit integers using calculator
Explains how to deal with repeats and random numbers outside range (PI by **both** 'different' numbers and 'numbers from 1 to 3200' seen).	AO2.4	E1	Ignore repeats and any (random) numbers outside the range Continue until 60 different numbers have been identified and select the students given those numbers
Explains how to select the 60 (expresses idea of matching numbers to students or selecting them)	AO2.4	E1	
			Total 4 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	States opportunistic (sampling). Accept opportunity/convenience.	AO1.2	B1	Opportunistic sampling
(ii)	Explains that sample is not random.	AO3.5b	E1	The sample is not random.
(b)	States both hypotheses correctly for one-tailed test	AO2.5	B1	H_0: $\mu = 66.5$ H_1: $\mu < 66.5$
	Formulates the test	AO1.1a	M1	

statistic			$z = \dfrac{65.4 - 66.5}{21.2/\sqrt{750}}$
Obtains the correct value of the test statistic	AO1.1b	A1	$= -1.42$
States the correct critical z-value OE	AO1.1b	B1	Critical z value = −1.28 −1.42 < −1.28
Infers H_0 rejected CSO	AO2.2b	A1	Reject H_0 − there is sufficient evidence that the advertising campaign has reduced the consumption of chocolate.
Correctly concludes in context. (FT only available if first B1 and M1 scored).	AO3.2a	E1F	
			Total 8 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Sets up enumerated population	AO3.1b	B1	Give each customer a unique number from 1 to 750.
Explains how enumerated population will be used to obtain sample with respect to random numbers	AO2.4	E1	Generate random integers from the calculator.
Explains how to deal with repeats	AO2.4	E1	Ignore repeats.
Explains how to identify 50 customers	AO2.4	E1	Continue until 50 different numbers have been identified and select the corresponding customers.
			Total 4 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Obtains probability from calculator	AO3.4	B1	$P(X \le 2) = 0.678$
(ii)	Obtains either of these figures (0.8791, 0.1074) PI	AO3.4	M1	$P(X \le 3) = 0.8791$ $P(X = 0) = 0.1074$
	Obtains correct probability	AO1.1b	A1	$P(1 \le X \le 3) = 0.8791 -$

				0.1074 = 0.772
(b)(i)	Recalls correct name for sampling method	AO1.2	B1	Opportunity sampling
(ii)	States that sampling method is unrepresentative giving one appropriate weakness	AO3.5b	E1	The 25 students all come from the same college and cannot be said to fairly represent all students. There could be a regional difference in diet.
(iii)	States both hypotheses using correct notation	AO2.5	B1	H_0: $p = 0.2$ H_1: $p > 0.2$
	States or uses B(25, 0.2) PI	AO3.3	M1	Under H_0 , use $X \sim$ B(25, 0.2) (where X represents number of students eating 5 or more portions)
	Obtains correct probability	AO1.1b	A1	P($X \geq 8$) = 0.109
	Evaluates model by comparing P($X \geq 8$) with 0.05 (condone 0.0468/0.047 used instead of 0.109)	AO3.5a	M1	0.109 > 0.05
	Infers H_0 accepted	AO2.2b	A1	Hence accept H_0
	States correct conclusion in given context	AO3.2a	E1	No significant evidence that more than 20% eat at least five a day
				Total 11 marks

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Circles correct answer	AO1.2	B1	The 2065 households in the village
(b)	Circles correct answer	AO1.2	B1	Simple random
				Total 2 marks

Q6.

Parameter is a numerical property of a population.

Parameter – population

E1

Statistic is a numerical property of a sample

Statistic – sample

E1

Numerical – all marks may be implied by examples

E1

3

[3]